

General

AA-Omniscience

- Description: AA-Omniscience is a knowledge and hallucination benchmark that measures factual reliability, rewards precise knowledge, and penalizes incorrect guesses or hallucinations. It provides a detailed assessment of a model's ability to distinguish known from unknowns across diverse knowledge domains.
- Paper: <https://arxiv.org/abs/2511.13029>
- Dataset: <https://huggingface.co/datasets/ArtificialAnalysis/AA-Omniscience-Public>
- Implementation:
 - The benchmark consists of 6,000 questions covering 42 topics, including *Business, Humanities and Social Sciences, Health, Law, Software Engineering, and Science, Engineering and Mathematics*.
 - Models are scored using the *AA-Omniscience Index*, which assigns points for correct answers, subtracts points for hallucinated responses, and keeps abstentions neutral, rewarding abstentions over incorrect guesses
 - Each answer is graded as either `CORRECT`, `INCORRECT`, `PARTIAL_ANSWER`, or `NOT_ATTEMPTED` based on the model's response and the ground truth answer. GPT-5.6 Luna (medium) is used as the grading model
 - Intelligence Index Integration: AA-Omniscience contributes two components to the Intelligence Index: (1) *Accuracy* - the proportion of correct answers, weighted at 10% of the overall Index, and (2) *Non-Hallucination Rate* - calculated as 1 minus the hallucination rate, weighted at 5% of the overall Index (AA-Omniscience's 15% share).

GDP.pdf

- Description: Artificial Analysis' implementation of [Surge AI's GDP.pdf](#), a benchmark that tests whether models can reason over long, real-world professional documents and satisfy task-specific criteria.
- Paper: <https://arxiv.org/abs/2607.11192>
- Dataset: [surgeai/GDP.pdf](#)
- Evaluation set: 100 tasks across ten professional domains, grounded in 4,592 PDF pages and graded against 1,275 criteria. We attempt every task five times, so each model has a fixed denominator of 500 attempts.
- Document preparation and delivery: We prepare each source PDF with [LiteParse 2.5.0](#), with English OCR enabled. Every model receives the complete extracted text of every page. When supported by the endpoint, we send page images first in page order, followed by the task and complete extracted text in a single user message. Models without image input receive text alone. We render page images at 150 DPI and reduce them, to a 72 DPI minimum, when model context or provider payload limits require it. We may also convert opaque pages from PNG to JPEG. Where an endpoint caps how many images one request may carry, we combine pages into composite images, two pages per image first and up to four where the cap is tighter, each cell labeled with its page number. Past four pages per

image, images cover the leading pages only; the remaining pages remain in the extracted text. When either applies, the prompt tells the model that page images are composites and the page number where image coverage stops. If the images cannot fit within context or request-size limits after adaptation, we send the complete extracted text alone. We do not truncate or summarize the extracted text. Models answer in a single turn without browsing or tools.

Unlike the Surge AI implementation, we do not use API document input features. These are opaque to the user and sit above the model layer, so they can introduce variation from API product decisions and hosts rather than comparing models apples-to-apples.

- Judging: GPT-5.6 Luna Medium judges each criterion independently. Each call receives the task prompt, contestant answer, and one criterion, but not the source PDF or contestant identity. We accept a task grading only when every criterion has a verdict. We score errors, missing attempts, and terminal input failures as zero.
- Reported metrics:
 - All-pass is the headline metric: the share of all 500 attempts where every criterion passes.
 - Mean Pass is the secondary metric: we compute each attempt's criterion pass rate, then average it with equal weight across tasks and repeats.
 - Domain cuts use the same task-macro Mean Pass calculation. We do not report domain All-pass.
- Cost and speed scope: Published costs cover contestant model calls only and exclude judge calls, PDF preparation, and OCR. We estimate time per task from output-token usage and model output speed. It excludes judge calls, PDF preparation, and OCR, so it is not an end-to-end evaluation timing.

Differences from the Surge AI implementation

Artificial Analysis		Surge AI
Document input	Text extracted with LiteParse and OCR, plus page images for image-capable models	Raw PDF sent to the provider's document input
Judge	GPT-5.6 Luna Medium	Gemini 3.5 Flash

The task set is shared, but the document input and judge differ, so Artificial Analysis and Surge scores are not directly comparable.

[AA-LCR v1.1](#)

- Description: Evaluate long context performance through testing reasoning capabilities across multiple long documents (~100k tokens measured using cl100k_base tokenizer).

- Changes from AA-LCR: Adds a system prompt to clarify grading instructions, corrects 16 answer keys, and grades with GPT-5.6 Luna (medium). Scores are not directly comparable with v1.0.
- Dataset: <https://huggingface.co/datasets/ArtificialAnalysis/AA-LCR>
- Implementation:
 - 100 hard text-based questions spanning 7 categories of documents (Company Reports, Industry Reports, Government Consultations, Academia, Legal, Marketing Materials, and Survey Reports)
 - ~100k tokens (measured using cl100k_base tokenizer) of input per question, requiring models to support a minimum 128K context window to score on this benchmark. ~3M total unique input tokens spanning ~230 documents to run the benchmark (output tokens typically vary by model)
 - Model responses are evaluated using GPT-5.6 Luna (medium) as an equality checker with pass@1 scoring